# SOLUTIONS

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|---|---|---|---|
| Module: | Digital Signal Processing | | |
| Module Code | EBU5376 | Paper | B |
| Time allowed | 2hrs | Filename | **Solutions_2223_EBU5376_B** |
| Rubric | ANSWER ALL FOUR QUESTIONS | | |
| Examiners | | | |

## Question 1

a) This question is about the Laplace transform. **[12 marks]**

i) State the formula of Laplace transform on a continuous function $x(t)$. (2 marks)

ii) Perform Laplace transform to the following function

$$x(t) = 2\cos(3t)\,u(t).$$

(10 marks)

b) This question is about the z-transform, which is another transformation between real and complex domain. **[13 marks]**

i) State the relationship between the z-plane and the s-plane. (4 marks)

ii) Perform inverse z-transform to the following sequence

$$X(z) = \frac{-2z^2 - \frac{2}{3}z}{z^2 + \frac{2}{3}z - \frac{1}{3}}$$

(9 marks)

Answer:

a)

i) The Laplace transform of a signal $x(t)$ is defined as

$$X(s) = \mathcal{L}(x(t)) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$

[2 marks]



Solutions

ii)
We have
$X(s) = \mathcal{L}(x(t)) = \int_{-\infty}^{\infty} 2\cos(3t)\, u(t) e^{-st}\, dt$ [2 marks]

$$= 2\int_0^{\infty} \cos(3t)\, e^{-st}\, dt$$

$$= 2\int_0^{\infty} \frac{e^{3tj} + e^{-3tj}}{2} e^{-st}\, dt$$

[2 marks]

$$= \int_0^{\infty} e^{(3j-s)t} + e^{(-3j-s)t}\, dt$$

[2 marks]

We know that the Laplace transform of $e^{-at}u(t)$ is $\frac{1}{s+a}$ , hence

$$= \frac{1}{s-3j} + \frac{1}{s+3j}$$

[2 marks]

$$= \frac{s+3j+s-3j}{(s-3j)(s+3j)}$$

$$= \frac{2s}{s^2+9}$$

[2 marks]

b)
i) Straight lines in the s-domain [2 mark] corresponds to circles in the z-domain [2 mark]

ii)

$$X(z) = \frac{-2z^2 - \frac{2}{3}z}{z^2 + \frac{2}{3}z - \frac{1}{3}}$$

$$= \frac{-2z^2 - \frac{2}{3}z}{\left(z - \frac{1}{3}\right)(z+1)}$$ [2 mark]

$$= A\frac{z}{z - \frac{1}{3}} + B\frac{z}{z+1}$$

Therefore,

$$-2z - \frac{2}{3} = A(z+1) + B(z - \frac{1}{3})$$

For $z = -1,\ 2 - \frac{2}{3} = B\left(-1 - \frac{1}{3}\right)$

Solutions

$$\frac{4}{3} = -\frac{4}{3}B$$

$$B = -1$$

For $z = \frac{1}{3}$, $\quad -\frac{2}{3} - \frac{2}{3} = A\left(\frac{1}{3} + 1\right)$

$$-\frac{4}{3} = \frac{4}{3}A$$

$$A = -1$$

Hence, $X(z) = -\frac{z}{z+1} - \frac{z}{z-\frac{1}{3}}$ [4 marks]

$$x[n] = -(-1)^n u[n] - \left(\frac{1}{3}\right)^n u[n]$$ [3 marks]

Solutions

**Question 2**

This question is about the family of Fourier Transform, which consists of Fourier Series (FS), Fourier Transform (FT), Discrete Time Fourier Transform (DTFT), and Discrete Fourier Transform (DFT).

a) State the assumptions on the input signals by the four members of the Fourier Transform family. **[8 marks]**

b) State and prove the time-shift property of Fourier Transform. **[8 marks]**

~~c) Compute the Fourier transform of the constant signal $x(t) = 1$. **[9 marks]**~~

Part c) is out of scope in this year

Answers:

a)

1. Fourier Series assumes the input signal to be continuous [1 mark] and periodic [1 mark]
2. Fourier Transform assumes the input signal to be continuous [1 mark] and aperiodic [1 mark]
3. Discrete Time Fourier Transform (DTFT) assumes the input signal to be discrete [1 mark] and aperiodic [1 mark]
4. Discrete Fourier Transform assumes the input signal to be discrete [1 mark] and periodic [1 mark]

b)

The time-shift property of Fourier Transform states

- Suppose $\mathcal{F}(x(t)) \overset{FT}{\leftrightarrow} X(\omega)$, then for any real number $a$, we have

$$x(t-a) \overset{FT}{\leftrightarrow} X(\omega)e^{-j\omega a}$$

[2 marks]

Proof:

$x(t-a) \overset{FT}{\leftrightarrow} \int_{-\infty}^{\infty} x(t-a)e^{-j\omega t}dt$ [2 mark]

$= \int_{-\infty}^{\infty} x(t')e^{-j\omega(t'+a)}dt'$ [1 mark]

$= e^{-j\omega a} \int_{-\infty}^{\infty} x(t')e^{-j\omega t'}dt'$ [1 mark]

$= X(\omega)e^{-j\omega a}$ [2 marks]



Solutions

**Question 3**

This question is about the Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT).

a) DFT is the only member of the Fourier Transform family that can be implemented by computers. **[8 marks]**

i) State the order of complexity of an N-point DFT. (2 marks)

ii) Perform DFT to the following sequence:

$$x[n] = [1, -j, -1 - j, 2j]$$

(6 marks)

b) FFT is the low-complexity algorithm for computing DFT. **[17 marks]**

i) State the order of complexity of an N-point FFT. (2 marks)

ii) Using a butterfly diagram, apply decimation in time (DIT)-FFT to the following sequence You are no longer required to memorize the butterfly diagram, it will be given in the question

$$x[n] = [1, -j, -1 - j, 2j]$$

(6 marks)

~~iii) FFT reduces the computational complexity of DFT by utilizing the divide and conquer approach. Show that an N-point DFT can be decomposed into the summation of two $\frac{N}{2}$-point DFTs.~~ Out of scope

~~(9 marks)~~

Answers:

a)

i)
$O(N^2)$ [2 marks]

ii)



Solutions

$x[n] = [1, -j, -1-j, 2j]$

$X[k] = \sum_{n=0}^{3} x[n] W_4^{nk}, \quad W_4^{nk} = e^{-\frac{j2\pi nk}{4}}$ [2 marks]

For k=0, $X[0] = \sum_{n=0}^{3} x[n] W_4^{n0} = x[0] + x[1] + x[2] + x[3] = 0$

For k=1, $X[1] = \sum_{n=0}^{3} x[n] W_4^{n1} = x[0] + x[1]W_4^1 + x[2]W_4^2 + x[3]W_4^3 = 1 + (-j)\cdot(-j) + (-1-j)\cdot(-1) + 2j\cdot j = 1 - 1 + 1 + j - 2 = -1 + j$

For k=2, $X[2] = \sum_{n=0}^{3} x[n] W_4^{n2} = x[0] + x[1]W_4^2 + x[2]W_4^4 + x[3]W_4^6 = 1 + (-j)\cdot(-1) + (-1-j)\cdot 1 + 2j\cdot(-1) = 1 + j - 1 - j - 2j = -2j$

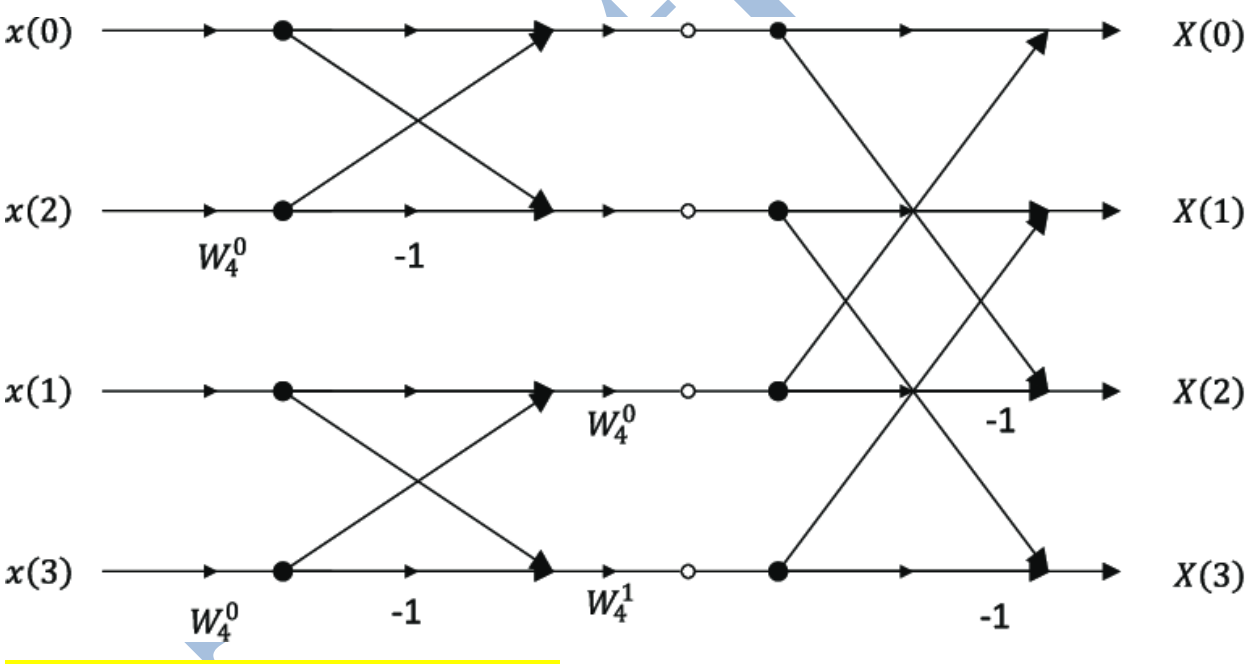
For k=3, $X[3] = \sum_{n=0}^{3} x[n] W_4^{n3} = x[0] + x[1]W_4^3 + x[2]W_4^6 + x[3]W_4^9 = 1 + (-j)\cdot j + (-1-j)\cdot(-1) + 2j\cdot(-j) = 1 + 1 + 1 + j + 2 = 5 + j$

Hence, the DFT of $x[n]$ is $X[k] = [0, -1+j, -2j, 5+j]$ [4 marks, 1 for each answer]

b)

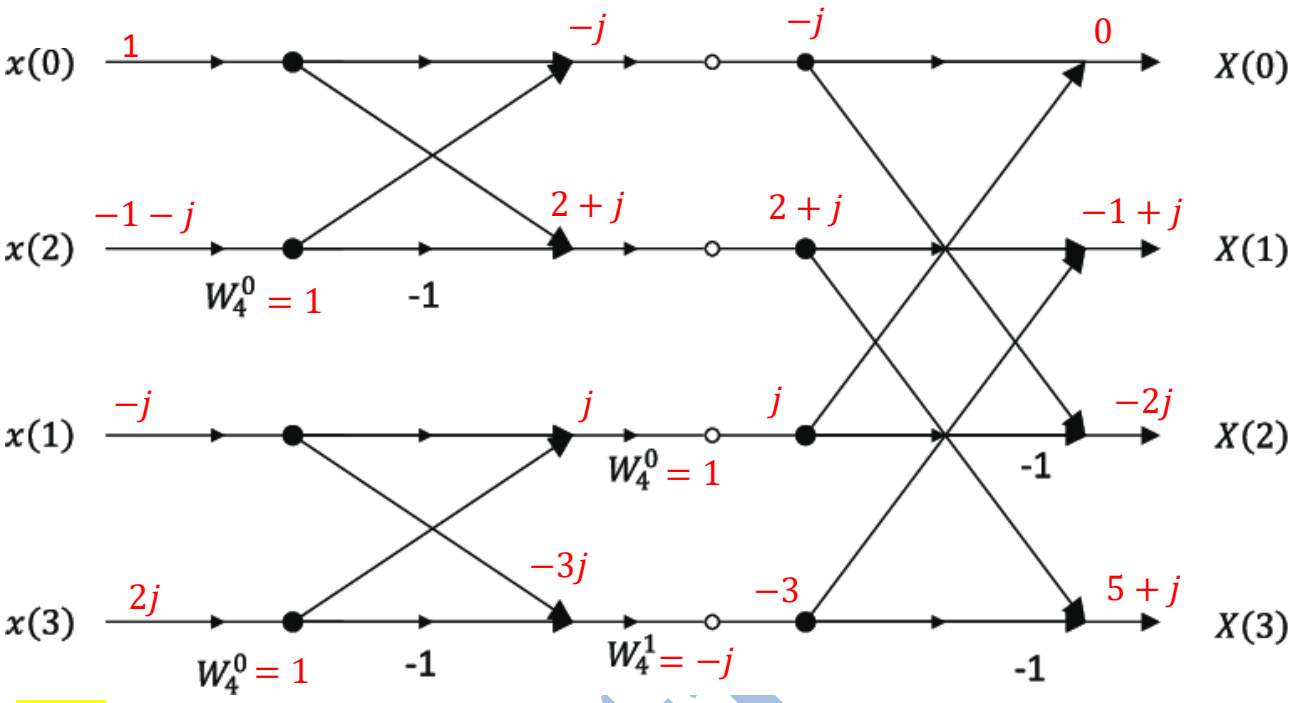
i) $O(N \log N)$ [2 marks]

ii)




[2 marks for correctly connecting the lines]
[1 mark for correctly labelling the input and outputs]

Then, the student needs to substitute the inputs and calculate the value at each node on the diagram, as shown in the image below.




[1 mark for the intermediate values]

The output is $X[k] = [0, -1+j, -2j, 5+j]$ [2 marks = 0.5 for each output]

Solut

**Question 4**

a) There are two types of digital filters, namely, Infinite Impulse Response (IIR) and Finite Impulse Response (FIR).

The Generic Digital Filter Transfer Function is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{i=0}^{M} b_i z^{-i}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

State and explain the condition for a filter to be FIR and comment on the stability of FIR filters. **[4 marks]**

b) The difference equation of a particular non-recursive filter is:

$$y[n] = 0.6x[n] + 0.5x[n-1]$$

**[8 marks]**

i) Draw its signal flow graph. (2 marks)

ii) Determine the filter's transfer function. (1 mark)

iii) Hence calculate the amplitude response at frequencies of 0, π/4 and π/2 radians. Full working must be shown. (5 marks)

c) The following transfer function is that of a recursive filter:

$$H(z) = \frac{(z+1)(z-1)}{(z-0.9)(z+0.6)}$$

**[13 marks]**

i) Draw its poles and zeros on the z-plane and comment on its stability. (8 marks)

ii) Determine the filter's difference equation. (5 marks)

Answer:

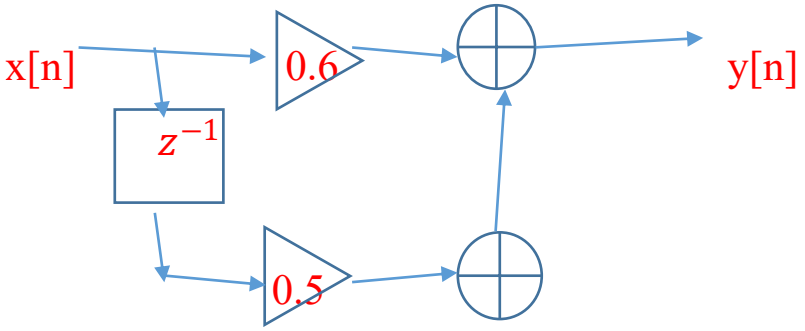
a)

$a_k = 0$ [1 mark] That is, the filter transfer function contains only zeros [1 mark] and the filter is non-recursive [1 mark]. All FIR filters are stable [1 mark].

Solutions

b)

i)




[2 marks]

ii)

$$y[n] = 0.6x[n] + 0.5x[n-1]$$

Transfer function: $H(z) = 0.6z^{-0} + 0.5z^{-1} = 0.6 + 0.5z^{-1}$

[1 mark]

iii)

$z = e^{j\omega}$, so frequency response is:

$H(e^{j\omega}) = = 0.6 + 0.5e^{-j\omega} = 0.6 + 0.5\cos\omega - j0.5\sin\omega$ [1 mark]

$$|H(e^{j\omega})| = \sqrt{(R_e^2 + I_m^2)}$$

Magnitude of $H(e^{j\omega}) = \sqrt{(0.6 + 0.5\cos\omega)^2 + (0.5sin\omega)^2}$ [1 mark]

For ω = 0, $|H(e^{j\omega})| = 1.100$

For ω = π/4, $|H(e^{j\omega})| = 1.016$ [3 marks: 1 for each correct value]

For ω = π/2, $|H(e^{j\omega})| = 0.781$.

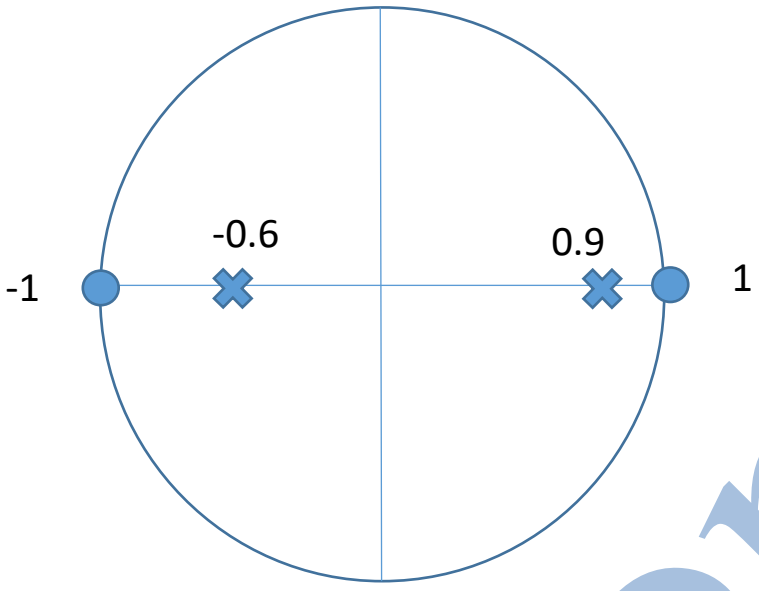
[NOTE: full working must be shown]

c)

Zeros at +1 and -1 [2 marks]

Poles at +0.9 and -0.6 [2 marks]

Solutions




[2 marks]

The filter is stable because both poles are inside the unit circle [2 mark]

ii)

$$H(z) = \frac{(z+1)(z-1)}{(z-0.9)(z+0.6)}$$

$$= \frac{z^2-1}{z^2-0.3z-0.54} = \frac{Y(z)}{X(z)} \quad [1\ mark]$$

$$= \frac{1-z^{-2}}{1-0.3z^{-1}-0.54z^{-2}} \quad [1\ mark]$$

$$Y(z)(1-0.3z^{-1}-0.54z^{-2}) = X(z)(1-z^{-2}) \ [1\ mark]$$

$$y_n - 0.3y_{n-1} - 0.54y_{n-2} = x_n - x_{n-2} \quad [1\ mark]$$

$$y_n = x_n - x_{n-2} + 0.3y_{n-1} + 0.54y_{n-2} \quad [1\ mark]$$

Solutions